Factors that Influence FIFA Player Market Value

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Table of Contents

Table of Contents **2**
Introduction **3**
Exploratory Data Analysis 3
Methods **4**
Market Value by Age Group 4
Market Value by International Ranking 5
Market Value by Overall Rating 6
Market Value by Nationality 7
Market Value by Position 7
Market Value by Preferred Foot 9
Conclusion **9**
Appendix **11**
Individual Reflections 11
Ralph Monroe 11
Anesu 11
Zoe 12
Vaishnavi 13
Kate 13
R Code 14

Introduction

Our dataset includes information on more than 18,000 professional football players from FIFA. Using this large player dataset in R, we explored how variables tied to physical traits, player performance, nationality and international reputation connect to market value across the global football market.

The dataset includes demographic variables such as age, nationality, club, position, height, weight, and preferred foot, along with performance measures such as overall rating, potential gap, dribbling, passing, finishing, and other technical skills. Financial variables such as market value, wage, and release clause were especially important because they directly reflect how players are valued economically.

From our investigation, we saw that overall player ratings and market value do not increase in a perfectly straight line. It gave us the ability to examine age and position differences, along with tactical factors such as whether specialized traits, like left footed players in certain positions, would influence the market value. To sum it all up, the dataset provided a strong foundation for looking beyond basic statistics and understanding what drives player value in today's football market.

Exploratory Data Analysis

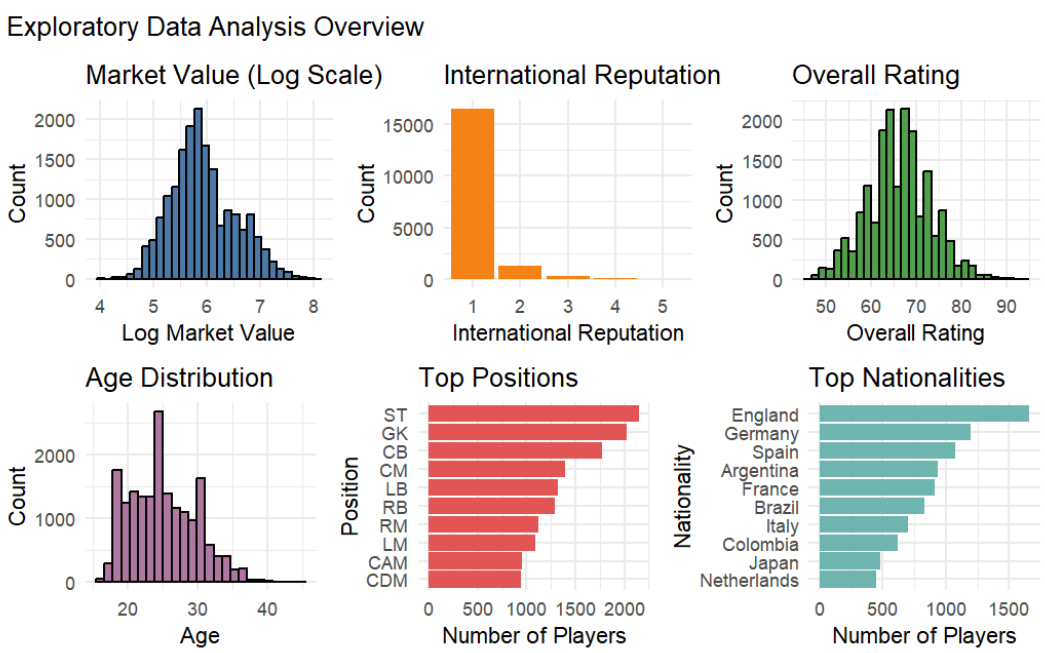

Figure 1

The exploratory data analysis (EDA) chart highlights key patterns across the dataset. We chose to illustrate the market value, the international reputation, overall rating, age, the top positions with their corresponding number of players, as well as the top nationalities and their player count. The log market value is right-skewed, with most of the players valued between €300,000 and €3 million. The international reputation has the highest count at level 1, indicating that most players have limited global recognition. The overall rating distribution is centered between 60 and 75, suggesting that the dataset is

dominated by mid-level players. From the age distribution, we can see that most players fall between 20 and 30 years old, reflecting the player's peak age. Additionally, we explored some positions, with the most common positions being striker (ST), center back (CB), and central midfield (CM). In terms of nationalities, we see the top 10 countries with the most represented ones being from England, Germany, and Spain.

Methods

Market Value by Age Group

To gain insight into a player's market value we initially used a scatter plot with the overall rating for the y-axis and the player's age on the x-axis. The market value was used for the dots on the scatterplot. However, with this dataset, the scatterplot did not present a trend due the extremely varied data. Therefore, we decided to use boxplots to see if they showed any patterns. We created two box plots using the same variables from the initial scatter plot (player age, overall rating, and market value). These two box plots were used to compare and gain better analysis into the market value of a player.

The first box plot looked if there were any patterns regarding the players age and overall rating. For age, we categorized players into 6 groups, which helped us observe how performance and market value can change when a player progresses through the different stages of their career. Furthermore, overall rating was an important variable. This rating is a combination of a player's abilities such as shooting, passing, defending, dribbling, and pace. Since the overall rating sums up a player's skill into a single number, it gives a strong indicator of the player's overall performance level. So, this is why we used overall rating to also help us compare performance across players.

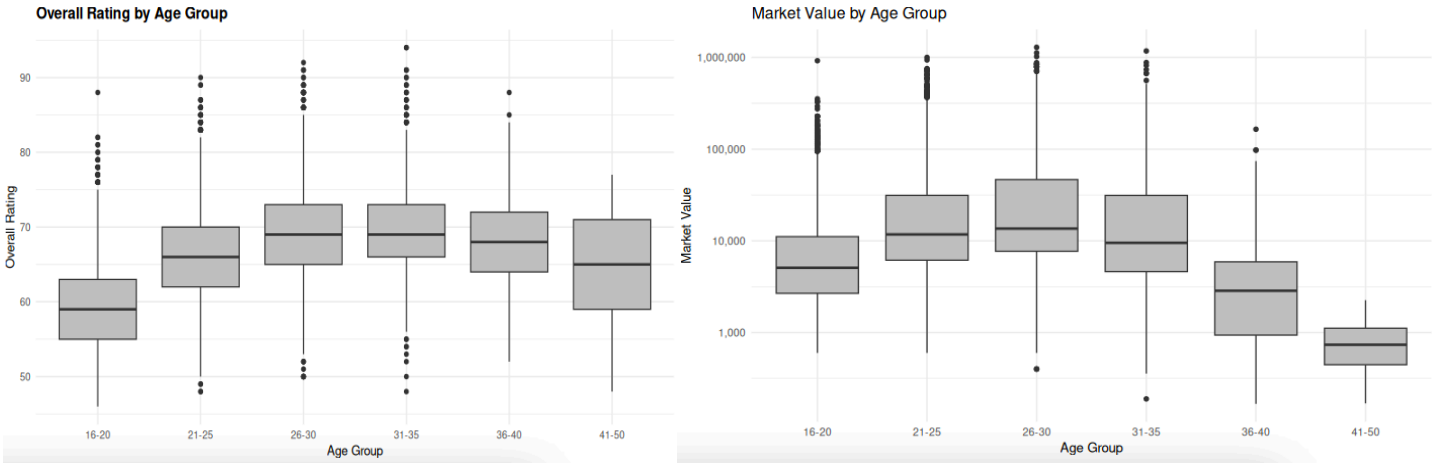

Figure 2

We used box plots for these visualizations because they were useful for showing the distribution across different categories. When we conducted our exploratory data analysis, we saw how the market value by rating changed across different age groups. There were clearly some outliers, and the majority of players

had overall ratings within the 60-75 range, but we were curious if this rating determined a player's market value.

Continuing with the box plot, we had to ensure the visualization was clear and easy to interpret. So, we had to use the logarithmic scale for the y-axis. Because the market values were varied, the log scale helped reduce the extreme high values without altering any data. What this visualization showed us was a clear pattern that most players have relatively low market values, while a small number of elite players have enormously high values, skewing the data. We observe that the 26-30 age group has the highest median market value, which suggests that players in this age range may be at their peak economic value. Alongside the age and overall rating boxplot, we see that this same age range (26-30) has the highest median overall rating. Thus, we can conclude that market value generally follows the same pattern as player performance. As players improve and reach peak skill levels in their late twenties, their market value also increases. Once players age beyond their peak, their market value tends to decline.

Market Value by International Ranking

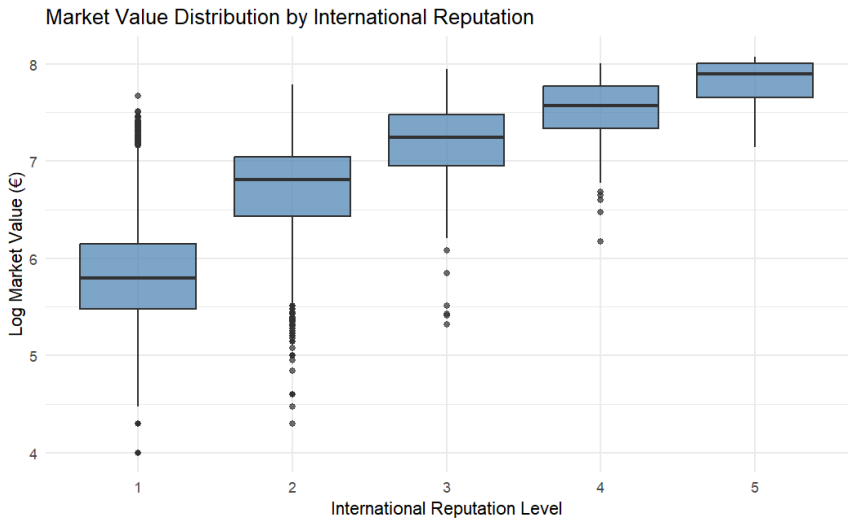

Figure 3

The boxplot shows log-transformed player market values across five levels of international reputation. These levels range from 1 to 5 where 1 is the lowest reputation and 5 is the highest. The international reputation is the player's global recognition and fame in the sport. By examining how market value distributions vary across these levels, we can identify if higher reputation is associated with higher market value.

From our visualization, we can see that there exists an upward trend in market value as international reputation increases. Players at level 1 have a median log market value ranging from log values of about 5.7 to 5.8 (€500,000 to €630,000). The interquartile range for this group is relatively wide, with most players falling between the 5.5 and 6.1 range (approximately €320,000 to €1.3 million), and several lower-value outliers being valued at log 4.0 (roughly €10,000). As the reputation increases to 2, the median market value rises to around 6.8 (approximately €6.3 million). The majority of players in this category fall between 6.4 and 7.0, which is equivalent to roughly €2.5 million to €10 million. At level 3,

the median increases further to an approximate log value of 7.2 (around €16 million), with most values ranging between log 7.0 and 7.4. The level 4 group continues this pattern, showing a median of near 7.5 (roughly €31 million).

Level 5 shows the largest market values overall with the median log market value ranging from 7.9–8.0 (approximately €80 million to €100 million). Compared with the lower reputation groups, this category displays higher central values and a tighter spread, indicating that players with the highest reputation consistently command very high market values and they are much fewer. The positive correlation communicates to the audience that the more notoriety a player has, the more they are going to be monetarily valued.

Market Value by Overall Rating

This visualization examines how a player's skill level, measured by their Overall Rating, relates to their market value. Players were grouped into eight rating brackets ranging from 46-55 at the bottom to 86+ at the elite end. Market value is plotted on a log scale, meaning each step up the y-axis represents a tenfold increase in value, and a polynomial curve is fitted to capture the overall trend smoothly across all sampled players.

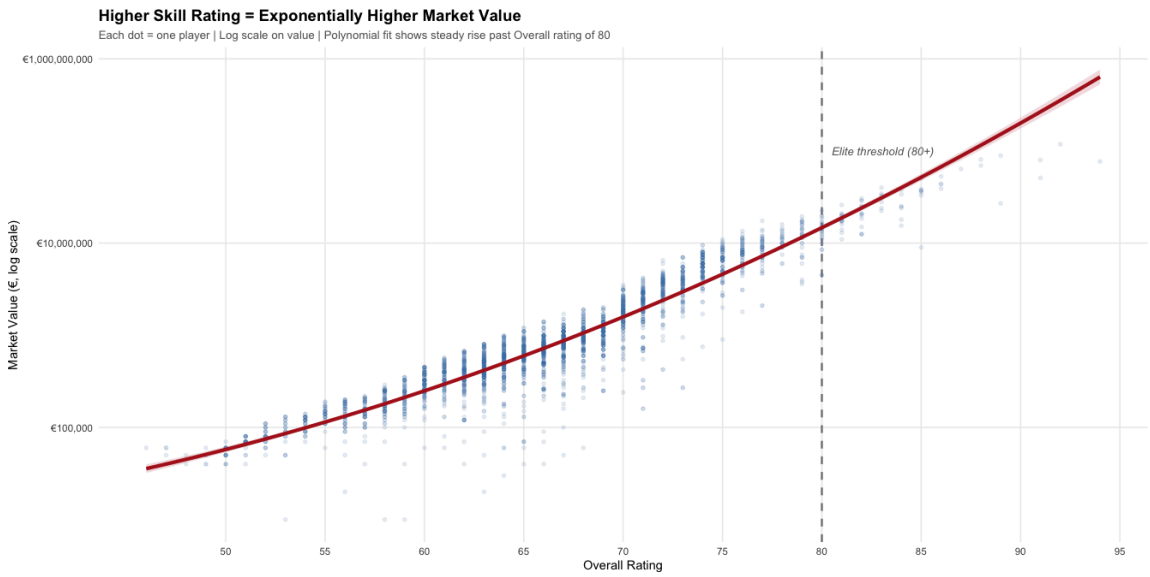

Figure 4

The results reveal a striking exponential relationship rather than a linear one. Players rated 46-55 have a median market value of just €0.1M, and this barely moves through the mid-tier brackets. A player rated 66-70, which is the most populated bracket with 5,087 players, is worth €0.92M. However, once a player crosses the 80 threshold the numbers change dramatically. Players rated 76-80 command a median of €9.5M, those rated 81-85 jump to €22.5M, and the elite 86+ bracket reaches €53.5M. This represents a 535-fold increase over the lowest bracket, driven by only 77 players occupying that tier. The gap between mean and median also widens at the top, with the 86+ bracket showing a mean of €55.27M, confirming that elite players consistently command exceptional prices. Notably, the polynomial curve steepens past the 80 threshold even on a log scale, meaning value is not just growing but accelerating faster than the

scale itself can compress it. The market is not simply rewarding improvement. It is rewarding irreplaceability, with a clear financial threshold sitting at an Overall Rating of 80.

Market Value by Nationality

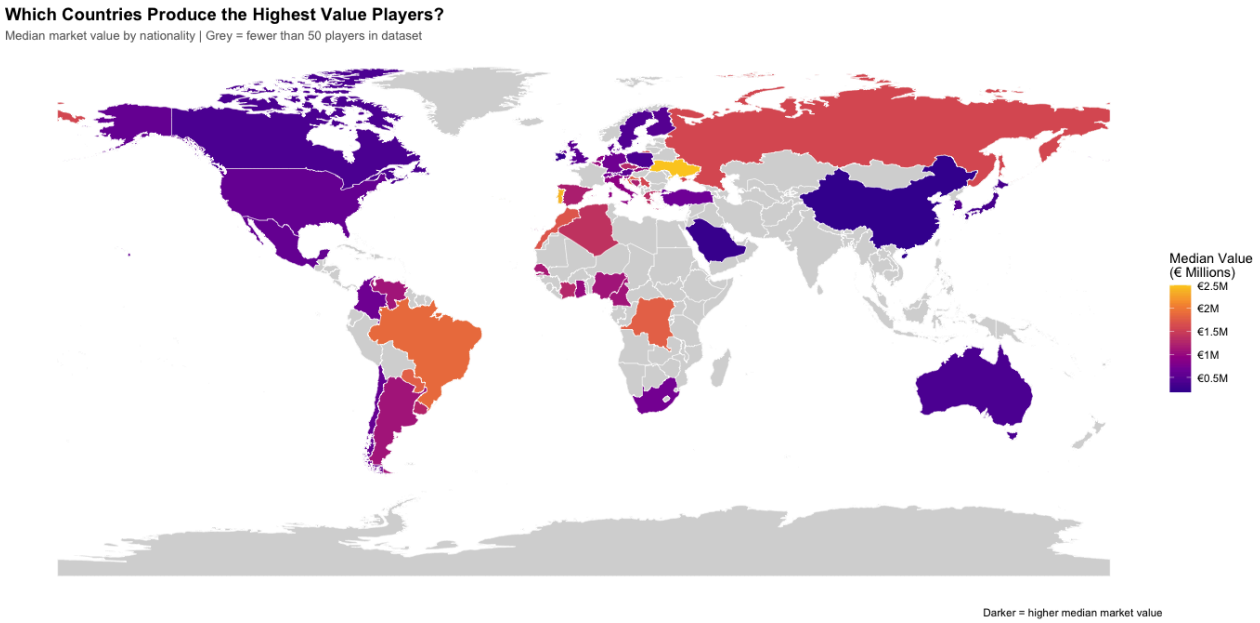

Figure 6

This visualization maps median player market value across nationalities, restricted to countries with at least 50 players in the dataset to ensure statistical reliability. The warmer colors (orange/yellow) represent higher median values, while the darker purple colors indicate lower median values, and the grey regions correspond to countries with fewer than 50 players. This visualization shows that countries such as Brazil and parts of Eastern Europe have higher median market values, while numerous nations across Africa and parts of Asia display lower median values. We observe that the European countries, North American regions, and Australia fall within moderate ranges. These patterns suggest that player market value is not evenly distributed globally, and from this we can infer that developmental issues in talent and facilities may be a cause of this. Overall, the visualization highlights how geography and national football ecosystems influence player valuation, providing important context for understanding disparities in the global football market economy.

Market Value by Position

Our analysis of the soccer market reveals a distinct financial ladder where value is driven by both tactical roles and specialized physical traits. By comparing values across the pitch, we identified a clear hierarchy: the closer a player is stationed to the opponent's goal, the higher their market value tends to be. Attackers and Midfielders clearly sit at the top of this structure, commanding both higher median values and much broader price ranges than their defensive counterparts.

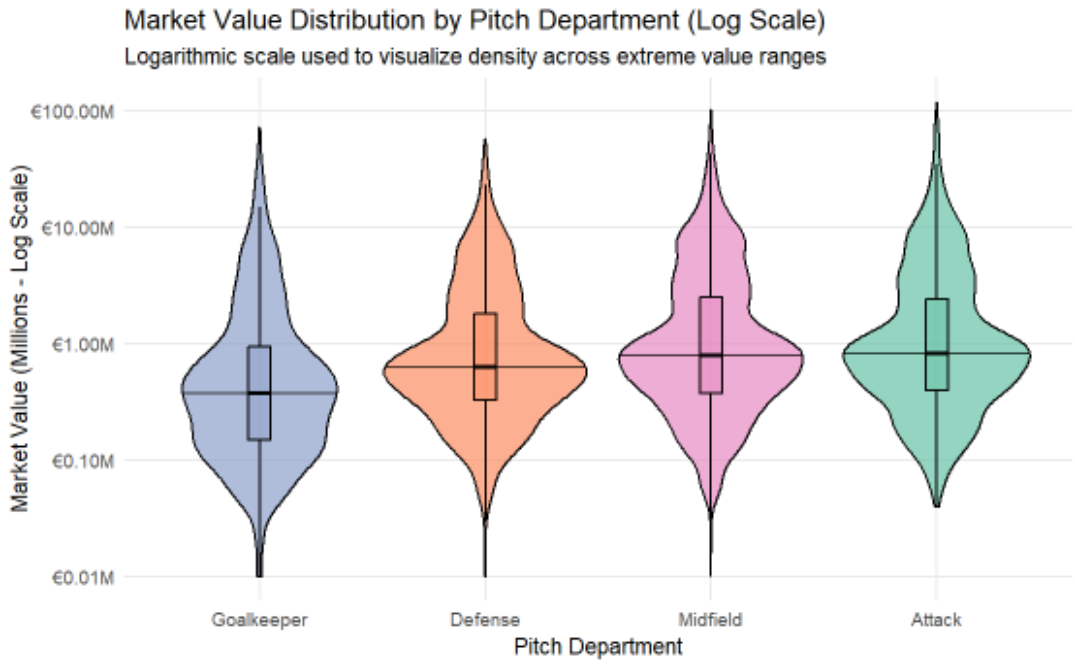

Figure 7

While defensive roles like GK (Goalkeeper) and CB (Center Back) are characterized by stability, with median values generally sitting below €1M, the market for offensive talent is far more volatile and expensive. The violin plot in figure 7 provides a unique look at player density across the main pitch departments. The wide base at the bottom of every shape indicates that the vast majority of professional players, regardless of whether they are Goalkeepers or Attackers are valued at under €2M.

From the long, thin tails that stretch upward toward the €12.5M+ mark, we can see the elite superstars of the football world. As we move from left to right across the departments, we see that the tails become longer. While the Goalkeeper and Defense clusters remain relatively "bottom-heavy," the Midfield and Attack shapes stretch significantly more. This proves that while most players are affordable, the financial "ceiling" for offensive talent is nearly unlimited compared to defensive roles.

Market Value by Preferred Foot

A critical finding in this deeper analysis is the "Scarcity Premium" associated with left-footed players in central and creative roles. In positions like LCM (Left Central Midfield), left-footed players do not just have a higher median value; they possess a vastly expanded valuation range, with top-tier talent reaching the €12M threshold because they are rare and unique. As seen in figure 8 on the next page, the trend persists in wide positions like RM (Right Midfield), thus reflecting demand for wingers who can create goal-scoring opportunities.

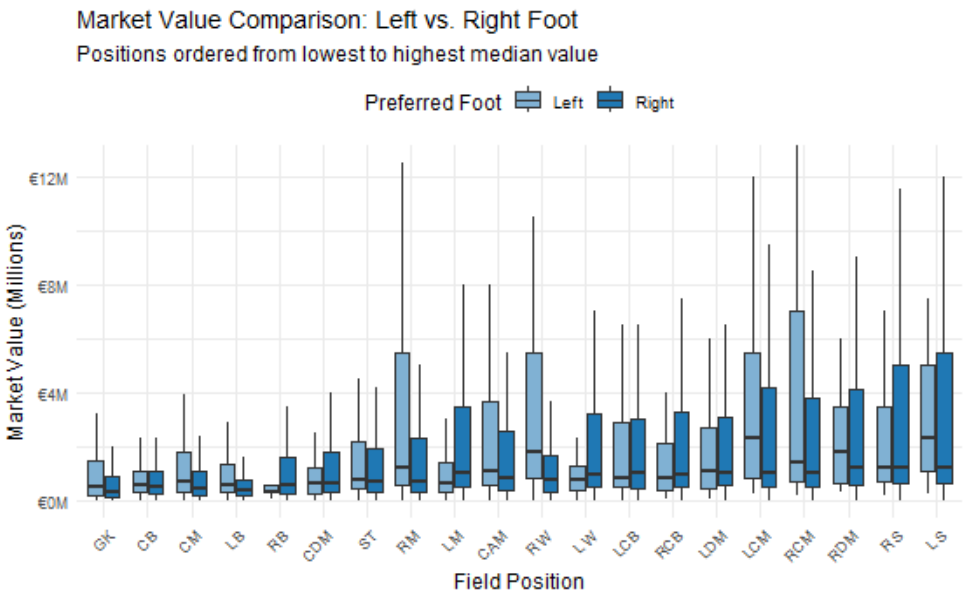

Figure 8

Conclusion

Our analysis shows that player market value in football is driven by several factors working together rather than one simple measure of talent. Performance remains one of the strongest drivers, but the data shows that value increases much faster once players move past the 80 overall rating range, where elite players begin to separate themselves financially.

Age also followed a clear pattern, with both market value and performance peaking between ages 26 and 30. International reputation showed a strong upward relationship with value, while position based results suggested that attacking and midfield roles generally carry higher market values than defensive roles. Geographic patterns also showed that Europe and South America continue to stand out in player value concentration.

Overall, the results suggest that proven performance, reputation, and positional demand play a much larger role in player valuation than physical traits alone. We can extend our study by gathering the data from players with similar traits and study them so that teams can figure out the best move they can make economically, where both the player and the organization can make a sizable amount of money.

Appendix

Individual Reflections

Ralph Monroe

This was a good project to work on and I appreciate each group member. I wanted to hit the ground running and find a dataset to work on, but after the first milestone we had to pivot to a new dataset. Once we found what we wanted to work on I shifted my focus to examining and exploring the FIFA data.

I found myself being fixated and working on the market value of a player's age. This included creating many visualizations that never saw the light of day. However, I was enamored with the many things R could do. It seemed like things were moving fast, so I stuck to the tried and true visualizations I've worked with before. So, I created box plots for the Market Value by Player Age and also Market Value by Overall Rating. I also sought to update the visualizations any time I heard feedback from our group meetings or the group messaging.

During the beginning of the group project I felt way more engaged. I tried to initiative and gather whatever information I could for the group and share it. Also, contributing to the coordination of our meetings. Towards the tail end of the project there was an uptick in work and I wasn't able to be as proactive as I wanted, but my group mates were amazing and ensured we were all on the same page.

As I reflect on this project, it was eye opening for me when it comes to data visualization. My day job is a Data Analytics Manager and I've seen countless visualizations and unused dashboards. What this project has taught me is to have a keen eye regarding visualizations. For example, just the other day, I was shown a dashboard that was overwhelming and the first thought I had was, this is chart junk!

The main thing I appreciate from this project is using R and seeing its visualization capabilities. Up until this point I stayed in my line of utilizing Python and SQL, but I was impressed by R's choropleth maps, the ease to scale data, and its vast libraries. Thus, I started looking into learning more about R and seeing how I can use it in my career. Overall, I enjoyed working on this project and being challenged to use skills I haven't used, learning new concepts, and working with cool people.

Anesu

This project was very rewarding as I learned a lot and got to figure out how to work with a team of people. I was the person in charge of communicating with the professor on behalf of the group. After getting feedback on our dataset, I then had to relay the message that we had to pivot from the Global Surface Temperatures to the Fifa dataset, as it had more angles that we could analyze from. Additional communication to the professor was done by attending class, emails and office hours.

I was more engaged in the later stages of the project and not so much in the beginning. I still cooperated but was really following instructions from other group members. In the later stages I did the final edits for

milestones 3,4 and the final report. I did the exploratory data analysis section, the visualization for the international reputation as well as some of the write up for the chloropleth map. I also tried to constructively give feedback for other visualizations and communicated the best I could the professor's feedback to the rest of the group. Towards the end I tried to collaborate and delegate tasks and the group responded well and positively, especially in the end the group was positive, quick and efficient with the feedback provided. Other than Milestone 4 which was difficult to coordinate the final edits due to some communication breakdown, communication for the final report went very smoothly.

In terms of the project, I learned more about appropriate color schemes and scales from the professor. I felt that my team was very creative and really showed me different ways to visualize the same data. They analyzed from angles I had not thought about before, especially when it came to a player's trade value and foot preference.

This project helped increase my proficiency in R and also helped me be more intentional about the graphs I might use for future research projects. I also learned how to participate as well as lead a team of future data scientists. I had no idea that something as simple as color can really change the full story of what the data is communicating. I hope to expand my knowledge in Tableau and SQL, as well as work with a group that has as great of a work ethic as this one.

Zoe

This project gave me a better understanding of how important teamwork and organization are when working through a full analysis. Because our group was large, there were times when it was harder to contribute in the exact way I originally expected, since tasks were divided across several people. Toward the end of the project, I stepped up more to help keep us organized, make sure we were staying on task, and fill in where needed so that no one person felt overloaded. I tried to support different parts of the project wherever I could, especially when deadlines got closer and we needed everything to come together clearly.

This project also connected closely to the kind of work I already do professionally, especially in how data needs to be prepared carefully before it can be used for reporting or decision making. At work, I often work with large datasets, review trends, and build summaries for others, so this project felt familiar in the sense that the goal was not only to run analysis but also explain the results clearly. It reinforced how important it is to think about the audience, because even when the code works, the findings still need to make sense to someone reading them. I also noticed that the same problem solving process applies in both settings: sometimes results are unclear at first, variables need to be checked again, or a better visual is needed before the pattern becomes clear. Overall, the project made me feel more confident combining technical analysis with interpretation, which is important both in class and in my job.

Vaishnavi

Coming into this project I was not the most familiar with football as a subject, but what drew me in quickly was how rich the dataset was and how many different angles you could explore from a single CSV file. My contributions focused on three visualizations centered on market value: Overall Rating, Potential Gap, and Nationality, and each one pushed me in a different direction.

For the Overall Rating visualization, I went in assuming the relationship between skill and value would be fairly straightforward. What I did not expect was just how dramatic the jump would be at the elite end. The most common player in the dataset sits in the 66-70 rating bracket with over 5,000 players worth a median of €0.92M, but just 77 players in the 86+ bracket command €53.5M each. Seeing that gap in the data was a genuinely surprising moment and it pushed me to think carefully about how to present it. A scatter plot with a loess curve ended up telling that story far better than any summary table could have.

The Potential Gap visualization was the one that challenged my assumptions the most. I went in expecting that players with the most room to grow would command the highest prices. The data said the opposite. Players already near their ceiling hold the highest median values, while raw young talent with large upside actually sits at the bottom. It taught me that the market rewards certainty over possibility, and it was one of those moments where letting the data speak changed how I thought about the question entirely.

Working with the Nationality map was my biggest technical challenge in the project. It was my first time using spatial data in R, and getting the rnaturalearth package, ISO country codes, and the color gradient legend to all work together correctly took more trial and error than I expected. Ukraine coming out on top of the rankings ahead of Portugal and Brazil was not something I anticipated, and it reinforced that surface level assumptions about football nations do not always hold up against actual data.

This project pushed my R skills further than any coursework had before, particularly with ggplot2, patchwork, and spatial visualization. But the bigger takeaway for me was learning to think about visualization as communication. Every decision from color to chart type to how you label a finding is really a choice about what you want your audience to understand. That is something I will carry forward well beyond this class.

Kate

I joined this project because our team was collectively drawn to the sports industry, and I was happy to contribute by sourcing our dataset from Kaggle. Working with my groupmates was an incredibly smooth and cooperative experience, which made it easy to turn our shared interest into a finished study.

My research focused on whether a player’s position and preferred foot act as financial multipliers. I started with a simple question: is there a hidden bonus for being left-footed? During the process, I learned that effective visualization often requires narrowing your focus; my early drafts used a €30M scale that made most players look like unreadable dots. By "zooming in" on a €12M window, I was finally able to

reveal a clear financial ladder where attackers and midfielders command much higher prices than defenders and goalkeepers.

Using scatter plots, I discovered that skill isn't priced the same across the pitch.An attacker's value skyrockets much faster than a defender's for every point they improve in skill. My most exciting find was proving the "lefty" bonus: in central midfield, the best left-footed players cost significantly more than their right-footed counterparts. Later, I used Violin and Ridge plots to show the shape of the market, which revealed that while most players are affordable, attacking positions have long, thin tails representing rare, multi-million-euro superstars.

This course has been pivotal for my growth as a future quantitative user researcher. Coming from a Mechanical Engineering background, I was already comfortable with technical systems, but this project taught me how to translate messy data into a clear story. A major highlight was learning the Tidyverse in R; it was a total eye-opener to see how logically and efficiently data can be cleaned and organized. I also really appreciated the opportunity to use both R and Tableau throughout the project. I now know how to choose the right scales and chart types to show the soul of the data.

R Code

Market Value By Age Group (Box Plot)

```
library(tidyverse)
library(viridis)
library(scales)

fifa_eda <- read.csv("fifa_eda_stats.csv")

# making sure values are whole numbers
fifa_eda$Value_num <- fifa_eda$Value

fifa_eda$Value_num <- gsub("€", "", fifa_eda$Value_num)
fifa_eda$Value_num <- gsub(",", "", fifa_eda$Value_num)

# convert to millions
fifa_eda$Value_num <- ifelse(
        grepl("M", fifa_eda$Value_num),
        as.numeric(gsub("M", "", fifa_eda$Value_num)) * 1000000,
         fifa_eda$Value_num)

# convert to thousands
fifa_eda$Value_num <- ifelse(
        grepl("K", fifa_eda$Value_num),
        as.numeric(gsub("K", "", fifa_eda$Value_num)) * 1000,
        fifa_eda$Value_num)

# numeric conversion
```

```
fifa_eda$Value_num <- as.numeric(fifa_eda$Value_num)

# turning the value per rating into a whole number
fifa_eda$Value_per_Rating <- round(fifa_eda$Value_num / fifa_eda$Overall)

# box plots scaling
ggplot(fifa_eda, aes(x = Age_Group, y = Value_per_Rating, fill = Age_Group)) +
        geom_boxplot() +
        scale_fill_viridis(discrete = TRUE) +

  scale_y_continuous( labels = comma, breaks = seq( 0,
        max(fifa_eda$Value_per_Rating, na.rm = TRUE), by = 200000)) +
        labs( title = "Market Value per Rating by Age Group", x = "Age Group", y = "Value per Rating")
+ theme_minimal()

ggplot(fifa_eda, aes(x = Age_Group, y = Value_per_Rating)) +
        geom_boxplot() + scale_fill_viridis(discrete = TRUE) +
        scale_y_log10(labels = comma) +
        labs( title = "Market Value by Age Group", x = "Age Group", y = "Market Value") +
        theme_minimal()
```

Market Value By Overall Rating

```
set.seed(42)

df_sample <- df %>%

  filter(!is.na(Value_EUR), Value_EUR > 0, !is.na(Overall)) %>%

  sample_n(3000)


ggplot(df_sample, aes(x = Overall, y = Value_EUR)) +

  geom_vline(xintercept = 80, linetype = "dashed",

        color = "grey50", linewidth = 0.8) +

  annotate("text", x = 80.5, y = 100000000,

       label = "Elite threshold (80+)",

       hjust = 0, size = 3.5, color = "grey40", fontface = "italic") +

  geom_point(alpha = 0.12, size = 1.2, color = "steelblue") +
```

```
  geom_smooth(method = "lm", formula = y ~ poly(x, 2), se = TRUE,
              color = "firebrick", fill = "firebrick",
              alpha = 0.15, linewidth = 1.5) +
  scale_y_log10(labels = label_dollar(prefix = "€", big.mark = ",")) +
  scale_x_continuous(breaks = seq(50, 95, by = 5)) +
  labs(title    = "Higher Skill Rating = Exponentially Higher Market Value",
       subtitle = "Each dot = one player | Log scale on value | Polynomial fit shows steady rise past Overall
rating of 80",
       x = "Overall Rating", y = "Market Value (€, log scale)") +
  theme_minimal() +
  theme(legend.position  = "none",
        panel.grid.minor = element_blank(),
        plot.title    = element_text(size = 14, face = "bold"),
        plot.subtitle = element_text(size = 10, color = "grey40"))
```

Market Value By Nationality

```
nat_value <- df %>%
  filter(!is.na(Value_EUR), Value_EUR > 0, !is.na(Nationality)) %>%
  group_by(Nationality) %>%
  summarise(Median_Value = median(Value_EUR, na.rm = TRUE),
            N_Players    = n()) %>%
  filter(N_Players >= 50) %>%
  mutate(ISO3           = countrycode(Nationality, "country.name", "iso3c"),
         Median_Value_M = Median_Value / 1e6)

world    <- ne_countries(scale = "medium", returnclass = "sf")
map_data <- world %>% left_join(nat_value, by = c("iso_a3" = "ISO3"))
```

```
p_map <- ggplot(map_data) +
  geom_sf(aes(fill = Median_Value_M), color = "white", linewidth = 0.2) +
  scale_fill_gradient(
    low      = "#c6dbef",
    high     = "#08306b",
    na.value = "grey85",
    name     = "Median Value\n(€ Millions)",
    labels   = function(x) paste0("€", round(x, 1), "M"),
    breaks   = pretty(nat_value$Median_Value_M, n = 5),
    guide    = guide_colorbar(
      barwidth     = 1,
      barheight    = 10,
      ticks        = TRUE,
      frame.colour = "grey50"
    )
  ) +
  labs(title    = "Which Countries Produce the Highest Value Players?",
       subtitle = "Grey = fewer than 50 players | Darker blue = higher median market value") +
  coord_sf(datum = NA) +
  theme_void() +
  theme(plot.title      = element_text(size = 13, face = "bold"),
        plot.subtitle   = element_text(size = 9, color = "grey40"),
        legend.position = "right")


p_bar <- nat_value %>%
  slice_max(Median_Value_M, n = 15) %>%
```

```
  mutate(Nationality = reorder(Nationality, Median_Value_M)) %>%
  ggplot(aes(x = Nationality, y = Median_Value_M)) +
  geom_col(fill = "#08306b") +
  geom_text(aes(label = paste0("€", round(Median_Value_M, 1), "M")),
            hjust = -0.1, size = 3.5,
            fontface = "bold", color = "grey30") +
  coord_flip() +
  scale_y_continuous(expand = expansion(mult = c(0, 0.25)),
                     labels = function(x) paste0("€", x, "M")) +
  labs(title    = "Top 15 Countries by Median Player Value",
       subtitle = "Minimum 50 players per country",
       x = "", y = "Median Market Value (€ Millions)") +
  theme_minimal() +
  theme(legend.position    = "none",
        panel.grid.minor   = element_blank(),
        panel.grid.major.y = element_blank(),
        plot.title    = element_text(size = 13, face = "bold"),
        plot.subtitle = element_text(size = 9, color = "grey40"))

p_map + p_bar +
  plot_layout(widths = c(2, 1)) +
  plot_annotation(title = "Market Value by Nationality",
                  theme = theme(plot.title = element_text(size = 15, face = "bold")))
```

Market Value By International Rank

```{r}
ggplot(fifa2, aes(x = factor(International.Reputation), y = log10(value_num))) +
  geom_boxplot(fill = "steelblue", alpha = 0.7) +
  labs(
    title = "Market Value Distribution by International Reputation",
    x = "International Reputation Level",
    y = "Log Market Value (€)"
  ) +
  theme_minimal()
```

Faceted Histogram

```{r}
ggplot(fifa2, aes(x = log10(value_num))) +
  geom_histogram(bins = 30, fill = "skyblue", color = "black") +
  facet_wrap(~ height_group) +
  labs(
    title = "Distribution of Market Value by Height Category",
    x = "Log Market Value",
    y = "Count"
  ) +
  theme_minimal()
```

Market Value By Position and Preferred Foot

```
# Load necessary libraries

library(tidyverse)

library(scales) # Essential for formatting currency (M/K) on chart axes

clean_currency <- function(x) {

  x <- str_replace_all(x, "[€]", "")

    # Handle Millions

  x <- ifelse(str_detect(x, "M"),

              as.numeric(str_replace(x, "M", "")) * 1000000,

              x)

    # Handle Thousands

  x <- ifelse(str_detect(x, "K"),

              as.numeric(str_replace(x, "K", "")) * 1000,

              x)

    return(as.numeric(x))

}

# Ensure 'fifa_eda_stats' is actually a data frame/tibble

fifa_clean <- fifa_eda_stats %>%

  mutate(

    Value_Numeric = clean_currency(Value),

    Wage_Numeric = clean_currency(Wage),

    Potential_Gap = Potential - Overall,

    # Using backticks for the space in the original column name

    International_Reputation = as.factor(fifa_eda_stats$International.Reputation)
```

```
  )
plot_data <- fifa_clean %>%
  filter(Position %in% common_positions)


# 2. Create the ordered boxplot
ggplot(plot_data, aes(x = fct_reorder(Position, Value_Numeric, .fun = median, .desc = FALSE),
                      y = Value_Numeric,
                      fill = Preferred.Foot)) +
  geom_boxplot(outlier.shape = NA) +
  # Using a focused 12M scale to see the boxes clearly
  coord_cartesian(ylim = c(0, 12500000)) +
  scale_y_continuous(labels = label_number(scale = 1e-6, suffix = "M", prefix = "€")) +
  scale_fill_manual(values = c("Left" = "#80b1d3", "Right" = "#1f78b4")) +
  labs(
    title = "Market Value Comparison: Left vs. Right Foot",
    subtitle = "Positions ordered from lowest to highest median value",
    x = "Field Position",
    y = "Market Value (Millions)",
    fill = "Preferred Foot"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "top") # Moving legend to the top gives the chart more width
```

```
fifa_clean <- fifa_clean %>%
  mutate(Dept = case_when(
    Position %in% c("ST", "LS", "RS", "LW", "RW", "CF") ~ "Attack",
    Position %in% c("CAM", "LAM", "RAM", "CM", "LCM", "RCM", "CDM", "LDM", "RDM", "LM",
"RM") ~ "Midfield",
    Position %in% c("CB", "LCB", "RCB", "LB", "RB", "LWB", "RWB") ~ "Defense",
    Position == "GK" ~ "Goalkeeper",
    TRUE ~ "Other"
  ))
cluster_plot_data <- fifa_clean %>%
  filter(Dept != "Other")


# 2. Create the Cluster Violin Plot
ggplot(cluster_plot_data, aes(x = reorder(Dept, Value_Numeric, FUN = median),
                              y = Value_Numeric, fill = Dept)) +
    geom_violin(trim = TRUE, alpha = 0.7, draw_quantiles = c(0.5), color = "black") +
    geom_boxplot(width = 0.1, color = "black", outlier.shape = NA, alpha = 0.5) +
    # Using trans_format and label_number to keep the axis labels in Millions/Euros
  scale_y_log10(labels = label_number(scale = 1e-6, suffix = "M", prefix = "€")) +
    scale_fill_brewer(palette = "Set2") +
    labs(title = "Market Value Distribution by Pitch Department (Log Scale)",
      subtitle = "Logarithmic scale used to visualize density across extreme value ranges",
      x = "Pitch Department",
      y = "Market Value (Millions - Log Scale)") +
```

```
  theme_minimal() +
  theme(legend.position = "none",
        panel.grid.minor = element_blank())
```

Exploratory Data Chart

```{r}
library(dplyr)
library(stringr)
library(ggplot2)
library(patchwork)

# --- Clean Market Value ---
fifa2 <- fifa2 %>%
  mutate(
    value_num = case_when(
      str_detect(Value, "M") ~ as.numeric(str_remove_all(Value, "[€M]")) * 1e6,
      str_detect(Value, "K") ~ as.numeric(str_remove_all(Value, "[€K]")) * 1e3,
      TRUE ~ as.numeric(str_remove_all(Value, "[€]"))
    )
  )

# --- Individual plots with clean labels ---
p1 <- ggplot(fifa2, aes(x = log10(value_num))) +
  geom_histogram(bins = 30, fill = "#4C78A8", color = "black") +
  labs(
```

```
    title = "Market Value (Log Scale)",
    x = "Log Market Value",
    y = "Count"
  ) +
  theme_minimal()

p2 <- fifa2 %>%
  filter(!is.na(International.Reputation)) %>%
  ggplot(aes(x = factor(International.Reputation))) +
  geom_bar(fill = "#F58518") +
  labs(
    title = "International Reputation",
    x = "International Reputation",
    y = "Count"
  ) +
  theme_minimal()

p3 <- ggplot(fifa2, aes(x = Overall)) +
  geom_histogram(bins = 30, fill = "#54A24B", color = "black") +
  labs(
    title = "Overall Rating",
    x = "Overall Rating",
    y = "Count"
  ) +
```

```
  theme_minimal()


p4 <- ggplot(fifa2, aes(x = Age)) +
  geom_histogram(bins = 25, fill = "#B279A2", color = "black") +
  labs(
    title = "Age Distribution",
    x = "Age",
    y = "Count"
  ) +
  theme_minimal()


p5 <- fifa2 %>%
  count(Position, sort = TRUE) %>%
  slice_head(n = 10) %>%
  ggplot(aes(x = reorder(Position, n), y = n)) +
  geom_col(fill = "#E45756") +
  coord_flip() +
  labs(
    title = "Top Positions",
    x = "Position",
    y = "Number of Players"
  ) +
  theme_minimal()
```

```
p6 <- fifa2 %>%
  count(Nationality, sort = TRUE) %>%
  slice_head(n = 10) %>%
  ggplot(aes(x = reorder(Nationality, n), y = n)) +
  geom_col(fill = "#72B7B2") +
  coord_flip() +
  labs(
    title = "Top Nationalities",
    x = "Nationality",
    y = "Number of Players"
  ) +
  theme_minimal()

# --- Combine ---
(p1 | p2 | p3) /
(p4 | p5 | p6) +
  plot_annotation(title = "Exploratory Data Analysis Overview")
```